

Assignment_Terraform_3

Terraform: Modules and Resources

1. What are modules in Terraform? How are they different from resources?

Modules in Terraform are reusable groups of Terraform configuration. A module can contain resources, input variables, outputs, and logic that together create one part of infrastructure. For example, a module can represent a complete VPC setup, an application server setup, or a database setup.

A resource is a single infrastructure object that Terraform manages, such as a virtual machine, security group, storage bucket, subnet, or load balancer. A module is bigger than a resource because it can group many resources together and make them reusable.

In simple words, a resource is one building block, while a module is a reusable design made from one or more building blocks. Resources create the actual infrastructure objects. Modules help organize those resources, reduce repeated code, and make the same setup easier to use in different environments.

2. How to create the modules and explain the steps in a brief way?

First, create a separate folder for the module. Many projects keep modules inside a folder named `modules`, and each module gets its own folder, such as `modules/network` or `modules/server`.

Second, add the module files inside that folder. Usually, `main.tf` contains the main infrastructure logic, `variables.tf` contains input values, and `outputs.tf` contains values that the main configuration may need after the module runs.

Third, keep the module flexible by using variables instead of hardcoding every value. This makes it easier to reuse the same module for development, testing, staging, or production.

Fourth, call the module from the main Terraform configuration by giving its source path and passing the required input values. After that, run `terraform init`, `terraform plan`, and `terraform apply` as usual.

Finally, test the module with a small use case before reusing it widely. A good module should be clear, reusable, and focused on one responsibility.